Avaliações

Pressure anisotropy on twin tunnel linings induced by a transverse gallery construction using time-dependent constitutive models

Comentário do revisor: The proposed abstract is aligned with the theme of the minisymposium.

This paper presents a thorough finite element study of stress redistribution in twin tunnels connected by a transverse gallery. The work is driven by the growing use of twin tunnels in urban infrastructure and the important role of gallery intersections, where load redistribution can threaten structural safety. Unlike traditional elastic analyses, this study considers the viscoelastic behavior of the concrete lining and the elastoplastic–viscoplastic properties of the rock mass, providing a more realistic long-term view. The methodology is carefully designed, incorporating shrinkage and creep effects in concrete (utilizing the Bažant–Prasannan solidification theory) and viscoplastic rock behavior based on the Drucker–Prager criteria. Model verification against previous results ensures reliability, while parametric simulations highlight the importance of rock friction angle and lining rheology. Results reveal significant pressure anisotropy at the tunnel–gallery intersection, with lateral zones experiencing up to twice the crown pressure. Viscoelastic linings help reduce long-term stresses by redistributing them, while higher rock friction angles decrease pressures and delay concentration effects. The paper’s main contributions are (i) demonstrating anisotropic pressure fields in twin tunnel linings under realistic long-term conditions, (ii) providing clear evidence of rheological effects in both rock and lining materials, and (iii) offering design insights for reinforcing junction zones. The writing is technically solid and generally clear, although some sentences are lengthy and a few some cases.

Typos and Corrections

Abstract

- “peak pressures are observed at the lateral sides of the walls” → *“peak pressures occur at the lateral sides of the tunnel walls.”*

Comentário do revisor:

- “offer valuable insights” → more formal: *“provide valuable insights.”*

Constitutive Models

- “stress–strain relationship under infinitesimal strain theory is expressed as” → *“...within the infinitesimal strain framework is expressed as.”*
- Beta_c in Equation (2) is not defined.
- “denotes, respectively, the plastic and viscoplastic multiplier” → *“denote, respectively, the plastic and viscoplastic multipliers.”*
- “i. e., cohesions cp and cvp assumed to remain constant” → *“i.e., cohesion parameters cp and cvp are assumed constant.”*
- Below Equation (4) \neta is used for the dynamic viscosity constant of the rock mass. It has been used before in Equation (2).

Spatial and Time Discretization

- “tunel-gallery” (Fig. 2 caption) → typo: *“tunnel-gallery.”*
- “after complete a predefined number of steps” → *“after completing a predefined number of steps.”*

Computational Model Verification

- “in this study were adopted fixed values” → *“in this study, fixed values were adopted.”*
- “Poisson ratio” → should be *“Poisson’s ratio.”*

Finite Element Simulations

- “a significantly increase is observed” → *“a significant increase is observed.”*
- “about twice and a half” → *“about two and a half times.”*
- “reach up to 25%” → *“reaching up to 25%.”*
- Figure 5 caption: “Logintudinal tunnel” → typo, should be *“Longitudinal tunnel.”*

Conclusions

- “the excavation of the transversal gallery generates” → *“the excavation of the transverse gallery generates.”*
- “in relation to the crown at φ = 0°” → smoother: *“relative to the crown at φ = 0°.”*

XLVI CILAMCE 2025

- Ref [1]: year appears truncated “202.” → should be *“2020s”* or correct year.